Goveronment College Women University Faisalabad

Department of Computer Science

Course title : Software Engineering

Course Code : CSC-415

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2541 (MA)

3rd Semester

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Project Name

Library Management System (LMS)

Project Background

Lately, managing books and records using physical register notebooks in our library has become a real hassle. Librarians spend too much time finding who took which book and when it is due for return. Often registers get misplaced or damaged over time, causing data loss. Also, students have to physically go to every shelf just to see if a specific book is available or already issued to someone else. To fix these daily issues, I selected this project to create a simple digital platform where all library records can be saved and checked easily on a screen instead of paper.

Objectives

- Replace manual paper registers with a digital platform to manage book records smoothly and accurately.
- Allow students to check whether a book is available or issued before visiting the library physically.

- Enable librarians to issue and receive books quickly with a few clicks without manual record-keeping.
- Automatically track return due dates and compute late fees to save time and eliminate manual calculation errors.

Stakeholders

- **Students:** Access the system to search for books, view their current issued book status, and check due dates for returns.
- **Librarian:** Handles daily library operations, such as adding new books, processing book issues/returns, and collecting late fines.
- **System Admin:** Oversees technical management, including user account creation, system updates, and database maintenance.

Requirements

Functional Requirements

- Secure login and account management for students, librarians, and administrators.
- A search bar allowing users to find books easily by title, author, or category.

- Instant display showing whether a specific book is currently available on the shelf or issued out.
- A digital interface for librarians to record book check-outs and returns efficiently.
- Automatic tracking of return deadlines and calculation of late submission fees.

Non-Functional Requirements

- A simple, clean, and intuitive user interface requiring no prior technical training.
- High responsiveness with search results loading within 2 to 3 seconds.
- Encrypted storage of user credentials and sensitive records within the database.
- Consistent data backups to prevent record loss or damage previously caused by physical registers.